

3	(a) force/load is proportional to extension/compression (provided proportionality limit is not exceeded)	B1	[1]
	(b) (i) $k = F/x$ or $k = \text{gradient}$	C1	
	$k = 600 \text{ N m}^{-1}$	A1	[2]
	(ii) $(W =) \frac{1}{2}kx^2$ or $(W =) \frac{1}{2}Fx$ or $(W =) \text{area under graph}$	C1	
	$(W =) 0.5 \times 600 \times (0.040)^2 = 0.48 \text{ J}$ or $(W =) 0.5 \times 24 \times 0.040 = 0.48 \text{ J}$	A1	[2]
	(iii) 1. $(E_k =) \frac{1}{2}mv^2$	C1	
	$= \frac{1}{2} \times 0.025 \times 6.0^2$		
	$= 0.45 \text{ J}$	A1	[2]
	2. (work done against resistive force $=$) $0.48 - 0.45 [= 0.03(0) \text{ J}]$	C1	
	average resistive force $= 0.030/0.040$	C1	
	$= 0.75 \text{ N}$	A1	[3]
	(iv) efficiency $= [\text{useful energy out/total energy in}] (\times 100)$	C1	
	$= [0.45/0.48] (\times 100)$		
	$= 0.94$ or 94%	A1	[2]